SAMBHAV SHRESTHA

sambhavshrestha111@gmail.com | New York, NY | linkedin.com/in/sambhav101 | sambhav101.github.io

EDUCATION

Stony Brook University – SUNY — August 2025 – June 2027
Master of Science, Computer Science
Relevant Coursework: Computer Vision, Natural Language Processing, Quantum Computing, Computer Architecture & Assembly, Operating Systems

St. Joseph's College New York — September 2018 – June 2022
Bachelor of Science, Computer Science and Mathematics — *GPA: 3.93*
Relevant Coursework: Advanced Algorithms & Programming, Advanced Databases, Multivariable Calculus, Linear Algebra, Probability & Statistics
Honors/Awards: SJC President's Scholar · Delta Epsilon Sigma Honor Society · Dean's List (4 semesters)

TECHNICAL SKILLS

Programming Languages: Python, Java, Kotlin, TypeScript, Ruby, C/C++, Rust, R, SQL, PostgreSQL, JavaScript, Verilog
ML / AI: PyTorch, TensorFlow, Hugging Face Transformers, CUDA, OpenCV, PyTorch Lightning, wandb, scikit-learn, RAG
Cloud & Infrastructure: AWS (EC2, Lambda, CloudWatch, SageMaker), Google Cloud Platform (Compute Engine, Kubernetes Engine, App Engine), Docker
Data & Backend: PostgreSQL, Flask, BeautifulSoup, ETL Pipelines, Pandas, NumPy, ggplot2, tidyverse, Shiny
Operating Systems: Linux / Unix, Windows
Certifications: Google Cloud Associate Cloud Engineer · Architecting with Google Compute Engine Specialization · Neural Networks and Deep Learning · Software Engineering Virtual Experience

PROFESSIONAL EXPERIENCE

HCL Technologies — March 2024 – July 2025
ML Infrastructure Engineer at Meta — *New York, NY*

- Architected and deployed **Sherlock**, a production RAG-powered internal assistant leveraging large language models to autonomously resolve infrastructure queries from engineers, research scientists, and cross-functional teams, substantially reducing escalation volume and accelerating onboarding to internal ML tooling
- Engineered comprehensive monitoring systems and a diagnostic doctor tool to track production LLM performance at scale, establishing real-time alerting pipelines for critical anomalies including entropy explosions, weight deviations, and throughput degradation
- Partnered directly with Research Scientists and ML Engineers to translate complex model requirements into robust infrastructure solutions, delivering end-to-end support across the full experimentation and deployment lifecycle
- Developed automated debugging tooling to detect and remediate errors across model packing, splitting, lowering, and transformation workflows, significantly reducing mean time to resolution for production incidents

Tarifica — May 2023 – February 2024
Software / Data Engineer — *New York, NY*

- Architected and maintained a fleet of over 300 web scrapers using Python and BeautifulSoup to systematically capture raw data from heterogeneous sources, transforming it into structured, analysis-ready records stored in PostgreSQL
- Designed and implemented scalable ETL pipelines using the Flask framework to validate, process, and route high-fidelity data to downstream analytics and visualization systems with strict accuracy and latency requirements
- Drove a comprehensive refactor of legacy codebases, improving throughput and output accuracy while introducing structured logging, observability instrumentation, and an automated test suite to ensure long-term production reliability

Amazon — July 2022 – March 2023
Software Development Engineer — *Seattle, WA*

- Designed, developed, and operated mission-critical microservices on AWS (EC2, Lambda, CloudWatch) powering the real-time checkout infrastructure for Amazon Go and Just Walk Out retail stores
- Spearheaded the end-to-end migration and modernization of a high-traffic service to AWS, delivering measurable reductions in operational cost, latency, and support ticket volume through enhanced observability, automated alarms, and fully automated CI/CD deployment pipelines
- Delivered production-quality features across Java, Kotlin, Python, TypeScript, and Ruby in a Linux environment, consistently meeting sprint commitments while upholding rigorous code quality and system performance standards

Microsoft – MSR Data Science Summer Institute (DS3) June 2021 – July 2021
Data Science Research Fellow *New York, NY*

- Extended the Financial Times investigative study on police complaints, conducting a rigorous statistical analysis of officer–victim race and gender dynamics across complaint datasets from NYC, Chicago, and Philadelphia
- Developed advanced regression and machine learning models in R to surface novel correlations and patterns in large-scale public records data, generating findings beyond those in the original publication
- Communicated results through publication-quality visualizations and formal presentations delivered to data scientists and engineers at Microsoft Research

Tarifica July 2021 – April 2022
Software Engineer Intern *New York, NY*

- Contributed to early-stage development of the data ingestion platform, building web scrapers and assisting in the design of the PostgreSQL schema for structured data storage
- Supported senior engineers in debugging and testing ETL pipeline components, gaining hands-on experience with production data workflows

ACADEMIC RESEARCH & PROJECTS

Argument Quality Ranking: LLMs vs. Fine-Tuned RoBERTa 2025
PyTorch · Hugging Face Transformers · scikit-learn · Python

- Conducted a systematic empirical study benchmarking fine-tuned RoBERTa against GPT-4.5, Llama 3, and Mistral on pairwise argument quality ranking across three difficulty levels and a held-out cross-topic generalisation set
- Developed RoBERTa v3 incorporating margin ranking loss and test-time pair flipping to eliminate positional bias, achieving 0.657 accuracy – matching GPT-4.5 (0.665) at a fraction of the inference cost; released model weights on Hugging Face (`SambhavSBU/argument-quality-roberta-v3`)
- Performed linguistic feature analysis revealing score gap as the dominant predictor of model correctness across all architectures, with hard pairs remaining near-random regardless of model scale or prompting strategy

Frequency Prior for Autoregressive Image Generation 2024 – 2025
PyTorch · CUDA · PyTorch Lightning · OpenCV · wandb

- Investigated the substitution of attention-based transformer blocks with Fourier-based frequency priors in autoregressive image generation architectures, targeting improvements in training stability, convergence speed, and perceptual output quality
- Designed and executed controlled ablation experiments to evaluate frequency-domain representations against attention baselines using quantitative image quality metrics

Pipelined MIPS Processor with Hazard Detection and Cache Integration 2024
Verilog

- Implemented a fully functional 32-bit five-stage pipelined MIPS processor in Verilog, incorporating forwarding, stall-based hazard resolution, and integrated instruction and data caches with verified correctness across memory-dependent instruction sequences

Police Complaints Data Analysis – Microsoft Research DS3 2021
R · ggplot2 · tidyverse · Bash

- Replicated and extended a published Financial Times investigation on police complaints, uncovering new statistical patterns in racial and gender distributions across complaint records from NYC, Chicago, and Philadelphia

INDEPENDENT PROJECTS

Stock Price Prediction Using LSTM *Python · TensorFlow · Matplotlib*

- Designed and trained an LSTM-based neural network for multi-step stock price forecasting, applying sliding-window preprocessing and dropout regularization to improve generalization

Digit Recognition Web Application *Python · TensorFlow · OpenCV · Flask · JavaScript*

- Built a full-stack web application backed by a trained deep CNN that classifies handwritten digits drawn on an in-browser canvas in real time with high accuracy

Twitter Sentiment Analysis Dashboard *R · Shiny · rtweet · ggplot2*

- Developed an interactive Shiny dashboard that retrieves live tweets by keyword and performs lexicon-based sentiment scoring with dynamically updating visualizations

Credit Card Fraud Detection *R · XGBoost · Random Forest*

- Evaluated and compared XGBoost, Random Forest, and Logistic Regression classifiers on a highly imbalanced fraud dataset, applying SMOTE oversampling and threshold tuning to maximize recall on positive cases

Minesweeper AI Agent *Python · pyGame*

- Built an autonomous Minesweeper agent using propositional logic inference and constraint propagation to identify safe cells, achieving near-optimal play on standard board configurations

Unbeatable TicTacToe Bot *Python · pyGame*

- Implemented a provably optimal TicTacToe adversary using the Minimax algorithm with alpha-beta pruning, guaranteeing a draw or win from any game state

LEADERSHIP & COMMUNITY

St. Joseph's College New York *Brooklyn, NY*
IT Student Technician January 2019 – June 2021

- Administered campus-wide IT support via an online ticketing system, triaging and resolving hardware, software, and network incidents for faculty and students
- Planned and executed AV and network infrastructure setup for campus-wide events in collaboration with the senior technician team

Peer Tutor, Mathematics & Computer Science January 2020 – May 2020

- Delivered one-on-one and small-group tutoring in calculus, statistics, and introductory programming, supporting student academic success across multiple course levels

RECOGNITIONS

Honors: SJC President's Scholar · Delta Epsilon Sigma National Honor Society · Dean's List (4 semesters)
Certifications: Google Cloud Associate Cloud Engineer · Architecting with Google Compute Engine Specialization · Neural Networks and Deep Learning (Coursera) · Software Engineering Virtual Experience